
UNCA Salaries and the Asheville Housing Market

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We examine UNCA Faculty and Staff Salaries in the light of proprietary cost of housing data obtained from [Mosaic Realty](#).

RESULTS OF THE TOTAL COMPENSATION TASK FORCE

UNCA's Total Compensation Task force [was charged](#) during academic year 2018/2019

to analyze current information and/or collect an additional underlying dataset on salary changes, cost of living metrics, and benefit expenses over time.

Ultimately, we chose the Southeast Consumer Price Index (or CPI) obtained through [The Bureau of Labor Statistics](#) as our sole cost of living metric and focused on the rate of growth of that metric as compared to the rate of growth in salaries and benefits here at UNCA. That allowed us to create graphs like the one shown in figure 1 showing that salaries at UNCA have generally grown just slightly faster than inflation.

While this comparison with CPI is useful - it is, by itself, rather unsatisfying. Many on the Faculty Senate complained that this clashes with their intuition that cost of living in Asheville has far outpaced the growth of salaries at UNCA and, more specifically, that housing costs are not accurately reflected in CPI. To be clear, the Task Force did attempt to consider these issues. We tried quite hard to get more data on cost of living in Asheville in general and on the housing market in particular. In addition to our own research, we consulted with the Asheville Chamber of Commerce but simply did not find reliable data on the housing market in Asheville.

I guess there's no point in giving up, though.

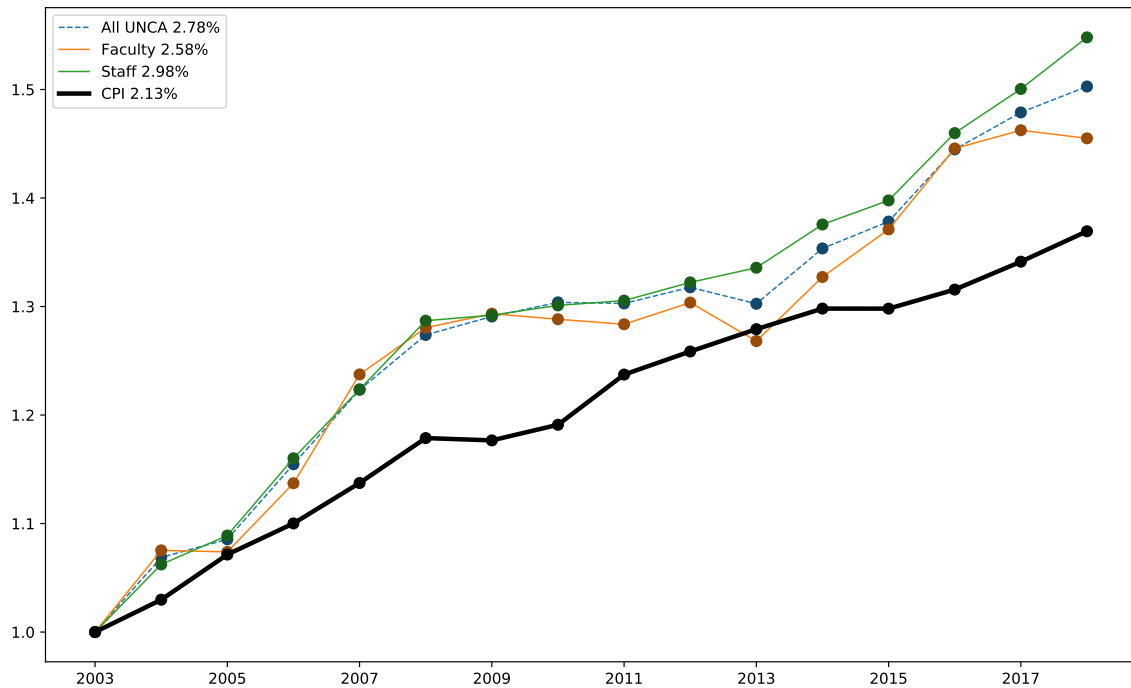


Figure 1: Rate of growth of salaries at UNCA compared to CPI

MOSAIC DATA

[Mosaic Realty](#) is a local Real Estate agency focusing on the immediate area. It is also a socially conscious company that donates broadly to local causes and, thus, is willing to share this data with me for the specific purpose of comparing housing data to UNCA salaries. The data is proprietary, however, and I am not at liberty to share the raw data files.

Mosaic has carefully curated data on homesales from [Carolina MLS](#) over the last 15 years. Thus, the analysis in this document is quite literally based on detailed information for every real estate transaction that has occurred in Buncombe County since 2003. When we refer to the median price of a house in a given year, we mean the median of the collection of all, actual sales prices during that year. We can also restrict the analysis by filtering the collection of houses on location.

With this data in hand, it's possible to forge ahead with the desired analysis. In figure 2, for example, we see the median sale price for homes in several areas - all of Buncombe County, the city of Asheville proper, and the neighborhoods of Montford and Oakley. The first two (the blue and aqua-blue lines in the middle of the graph) essentially reproduce Chart 3 on page 5 of [Mosaic's own market analysis report](#). With the full data at hand, though, we can drill down to get much more precise information - like the neighborhood curves for Montford and Oakley shown in the figure. Of course, the sample sizes tend to be smaller for neighborhoods and, thus, more volatile.

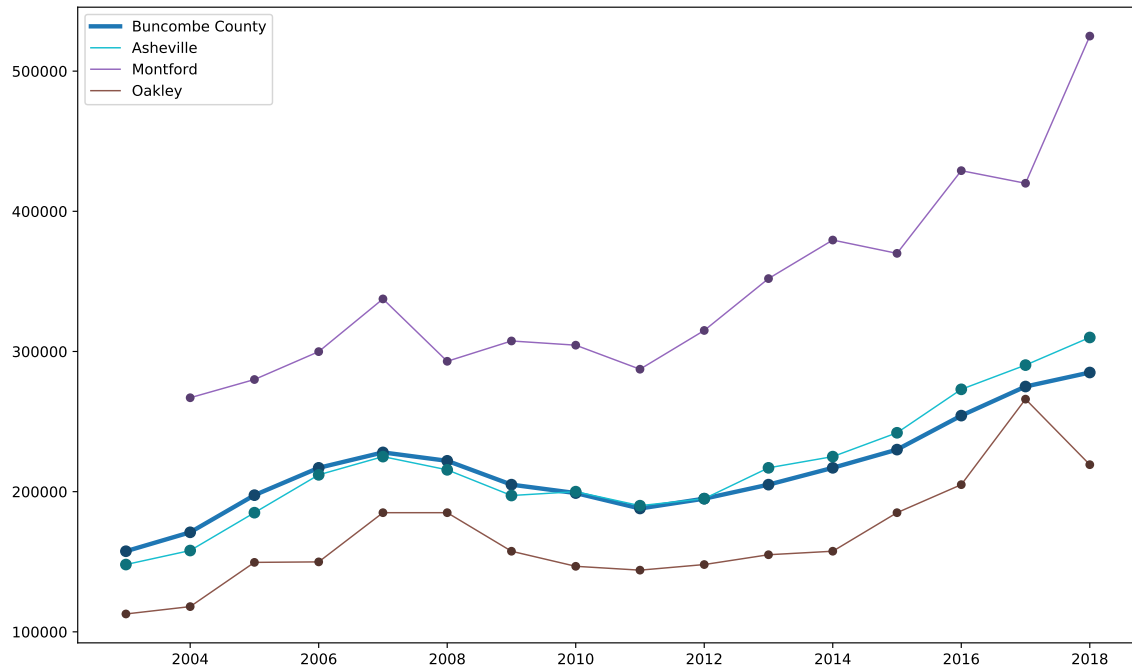


Figure 2: Median home prices for several areas

RATES OF GROWTH

A major objective of the task force was to compare the rate of growth of faculty and staff compensation to various living costs, as in figure 1. With this new data in hand, let's compare the rate of growth of all UNCA salaries to the rate of growth of housing prices. Let's also throw in the rate of growth of CPI to compare this to our old metric.

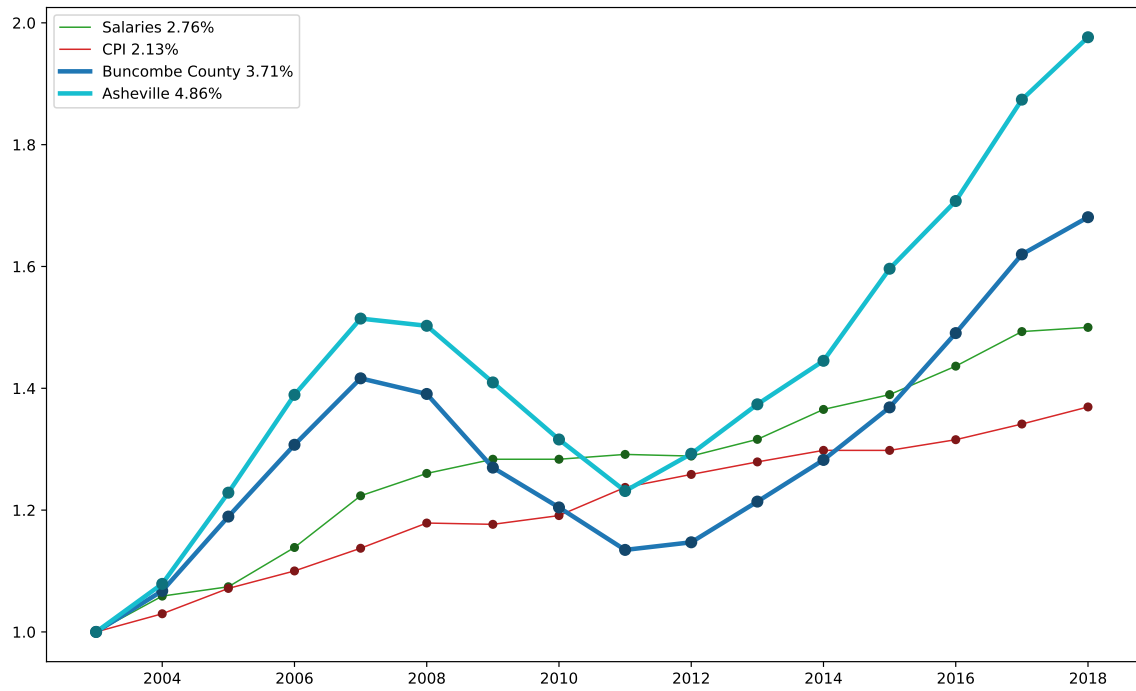
In order to compute the rate of growth of housing prices, I'm going to use the median price per square foot of all the homes sold in a particular region during years in the range of interest. We then divide by the median price in the first year to obtain a graph starting at 1 on the vertical axis. The resulting graph then shows, for subsequent years, how many times larger the median price per square foot is. We do a similar thing for all the variables that we wish to compare with and plot these together.

The result, going back as far as the data allows, is shown in figure 3a. A couple of things are clear from this picture:

- The average rate of growth of housing prices has, indeed, far outpaced the average rate of growth of UNCA salaries.
- Housing prices are generally much more volatile than salaries; housing prices took a much bigger hit during the economic downturn.

Figure 3b shows the relative growth since the recovery showing that, since then, the rate of growth of housing costs has been more than triple that of salaries.

(a) Since 2003



(b) Since 2011

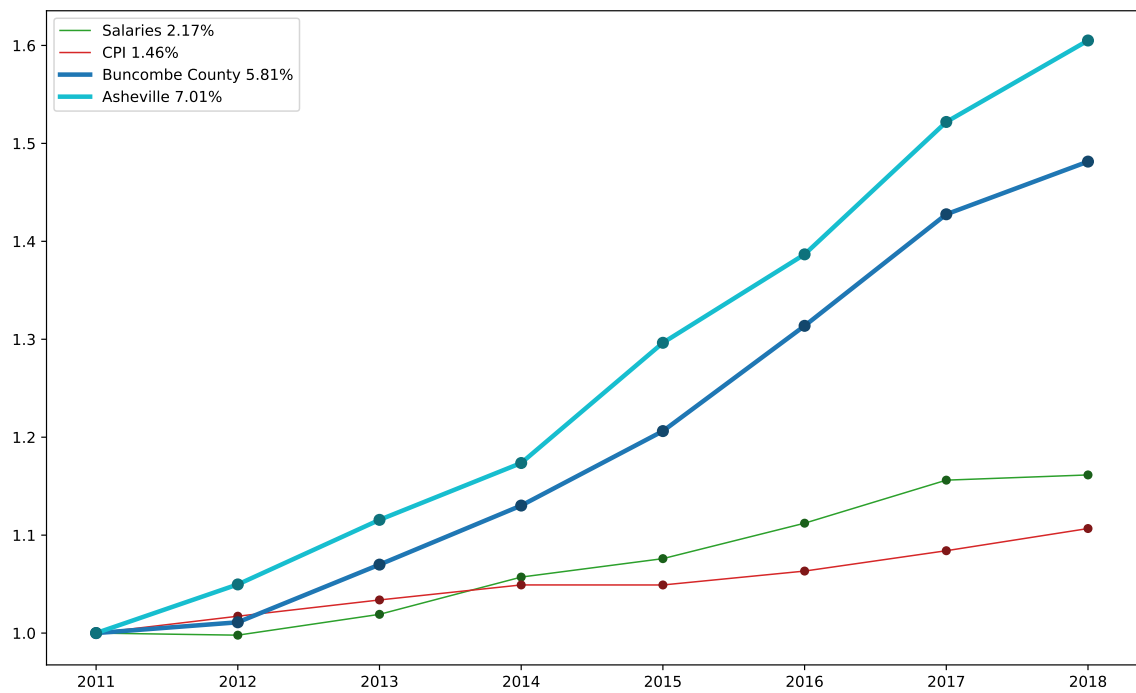


Figure 3: Salary growth compared to growth of housing prices

PURCHASING POWER

What effect has the change in income relative to housing prices over the last decade or so had on actual purchasing power? To be concrete, let's put ourselves in the place of a relatively new Assistant Professor looking for a home. Perhaps they are single or the primary earner in the household. They would like to live in the city limits because they've heard good things about Asheville City Schools but are otherwise flexible. What are the purchasing options available to this person relative to their counter parts a few years ago?

To examine this question, take a look at the graph shown in figure 4. The thick green line represents the median selling price of all houses sold within the Asheville city limits in any particular year. The thinner, dashed, green lines represent the corresponding quartiles - i.e. the 25th and 75th percentiles. Thus, the green lines taken together provide a picture of the range of the mid 50% of houses available.

Next, the red line in figure 4 is meant to be a rough measure of the purchasing power of the median assistant professor here at UNCA. It's computed very simply using a common rule of thumb, as espoused on [this web page by Fidelity](#) - namely, *the value of a house you purchase should be a maximum of 3 to 5 times your annual income, depending on your savings available for a downpayment*. In the ideal situation described there, a homebuyer has a full year's salary to put down on the house; such a homebuyer could be comfortable purchasing a house for 5 times their annual income. Homebuyer's who are already in debt should consider staying at 3 times their annual income. In order to approximate purchasing power for the typical Assistant Professor, I went right in the middle. Thus, the red line is simply 4 times the median salary of all Assistant Professors in a given year.

Given this interpretation, it's easy to see what's going on in figure 4. Back in 2003, the median assistant professor could afford within in the Asheville city limits if it was priced at the 58th percentile. Their purchasing power dipped below that level for a few years and then actually climbed up during the economic downturn. Since then though, it's decreased down to the point where they can only afford a house priced at the 40th percentile. We can even look at the houses clustered near these percentiles and examine their characteristics. For example, a house at the 58th percentile in the Asheville city limits in 2003 had over 1700 square feet while a house at the 40th percentile in 2018 had barely 1400 square feet.

Other case studies reveal similar decreases in purchasing power.

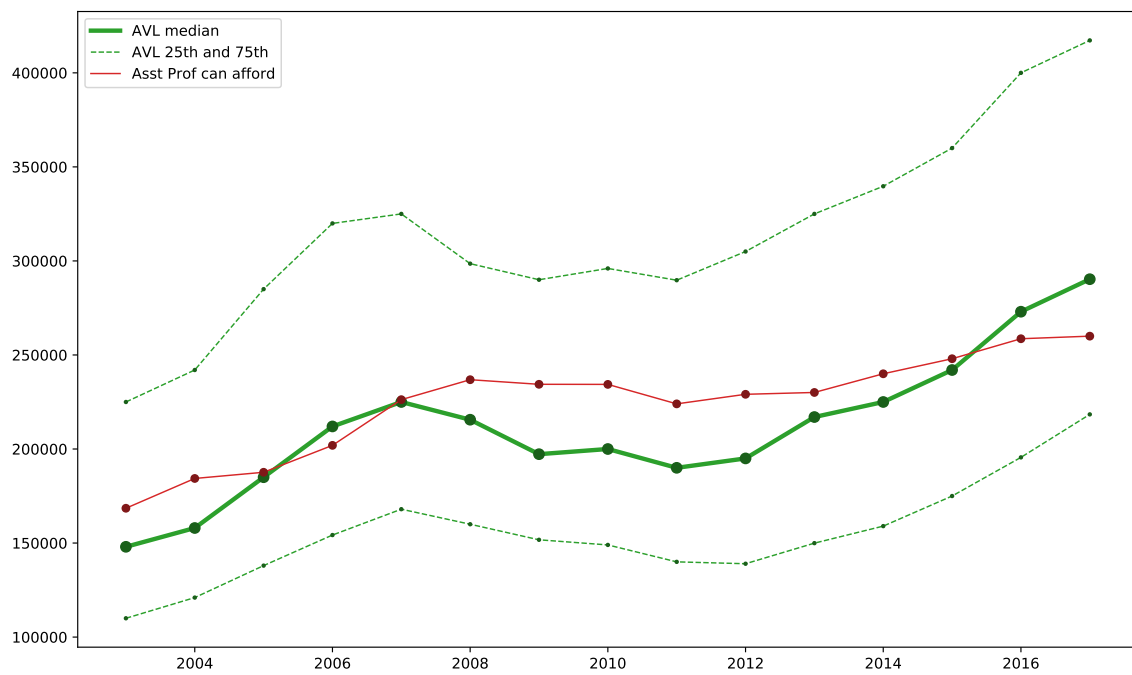


Figure 4: Asst prof purchasing power in Asheville proper